Netflix Content Analytics using Python, SQL & Data Transformation (ELT Pipeline)

Business Problem Statement

Netflix hosts thousands of movies and TV shows from different countries, genres, and directors. As the content library grows, it becomes increasingly difficult for business teams to understand content distribution, identify successful creators, evaluate genre performance, and make informed content acquisition or production decisions.

The objective of this project is to build an ELT pipeline that extracts raw Netflix data using an API, loads it into a Python(Pandas), and performs data transformation and business analysis using SQL. The analysis aims to uncover insights about directors, countries, genres, and content duration to support strategic content planning and business decision-making.

Business Objectives

The project focuses on answering the following business questions:

1. Identify directors who have successfully produced both Movies and TV Shows, helping Netflix recognize versatile content creators.
2. Determine which country has produced the highest number of Comedy Movies, enabling regional content performance analysis.
3. Identify the top-performing movie director each year (based on the date content was added to Netflix) to understand trends in content acquisition.
4. Analyze the average duration of movies across different genres to better understand audience content preferences and genre characteristics.
5. Find directors who have created both Comedy and Horror movies, highlighting creators with diverse filmmaking expertise.

Deliverables

Data Extraction

Extract data using an API.

Data Loading (Python)

Load it into the pandas and perform basic inspections and validations

Data Transformation (SQL)

Perform SQL based transformation & organize data into structured format, Data Cleaning, Null handling & Data normalizations etc